

NEOPLASIA

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Outlines

- Definition of neoplasia
- Epidemiology of tumors
- Causes of tumor
- Carcinogenesis
- Grading and staging of tumor
- Clinical aspect of neoplasms
- Prevention of cancers

EPIDEMIOLOGY

- Cancer is the **second leading cause of death** in the United States.
- In 2018, approximately 1,700,000 new cases of cancer were diagnosed in the United States and 600,000 people died from the disease.
- The most common cancers are **breast, lung, and prostate** in **descending order of frequency**

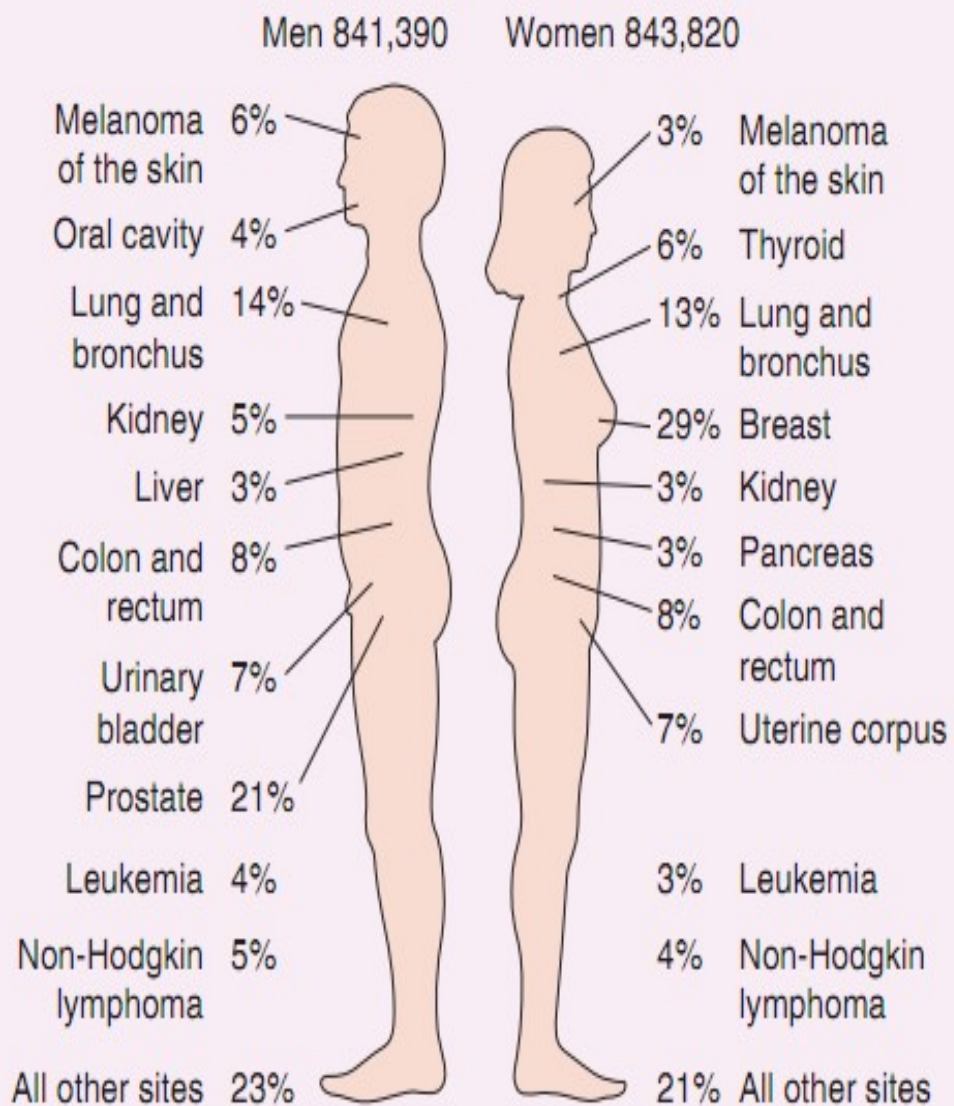
Cancer in Men

- In men, the sites with the highest cancer rates are (in order of decreasing frequency):
 - Prostate,
 - Lung and bronchus,
 - Colon and rectum
- The highest mortality rate-**lung** and **bronchus** cancers more commonly cause death than prostate cancer
 - Detected late in course
 - Limited screening method

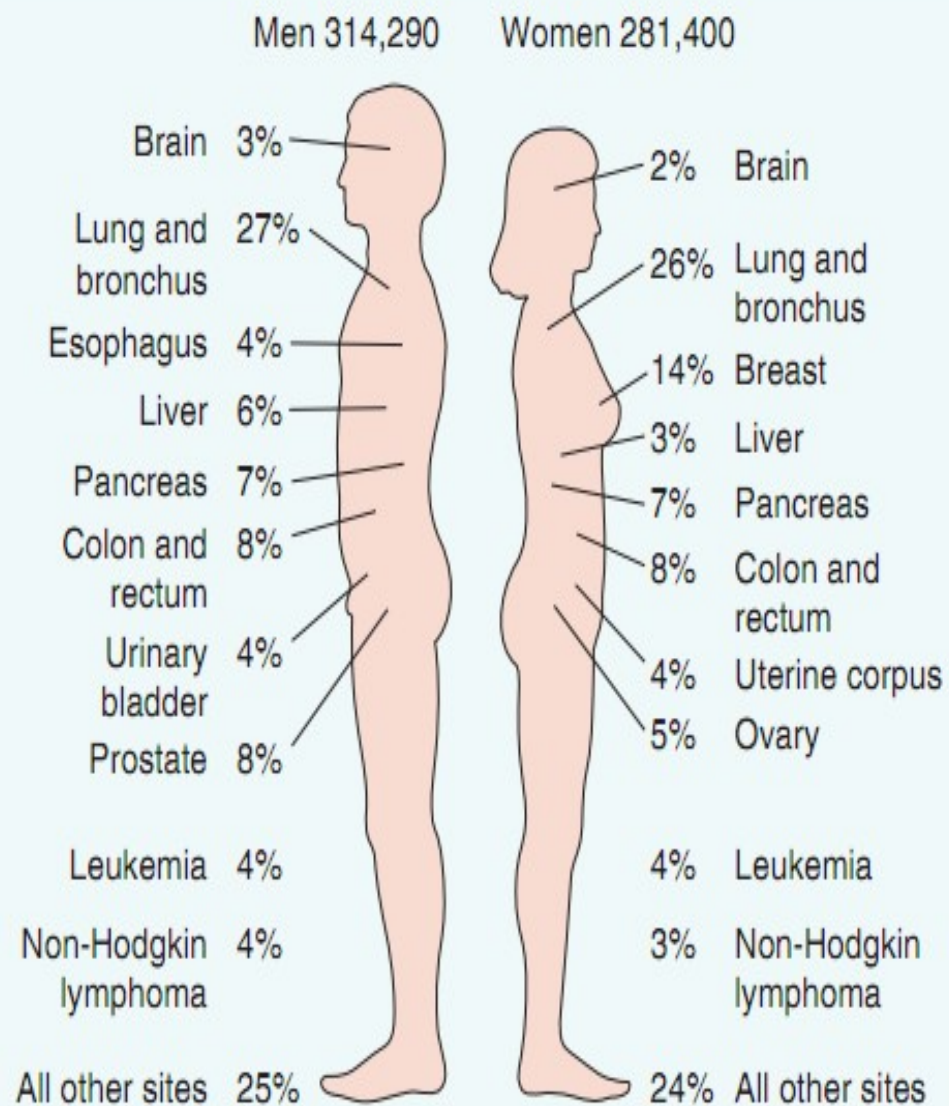
Cancer in Women

- In women, the sites with the highest new cancer rates are (in order of decreasing frequency):
 - Breast,
 - Lung and bronchus,
 - Colon and rectum
- Highest mortality rate- **lung** and **bronchus** cancers.

A 2016 ESTIMATED CANCER INCIDENCE BY SITE AND SEX*



B 2016 ESTIMATED CANCER DEATHS BY SITE AND SEX



Cancer in Children

➤ In children, the most common cancers are (in **descending** order)

- Hematologic malignancies (i.e. acute lymphocytic leukemia),
- CNS malignancy,
- Neuroblastoma, and
- Non-Hodgkin lymphoma

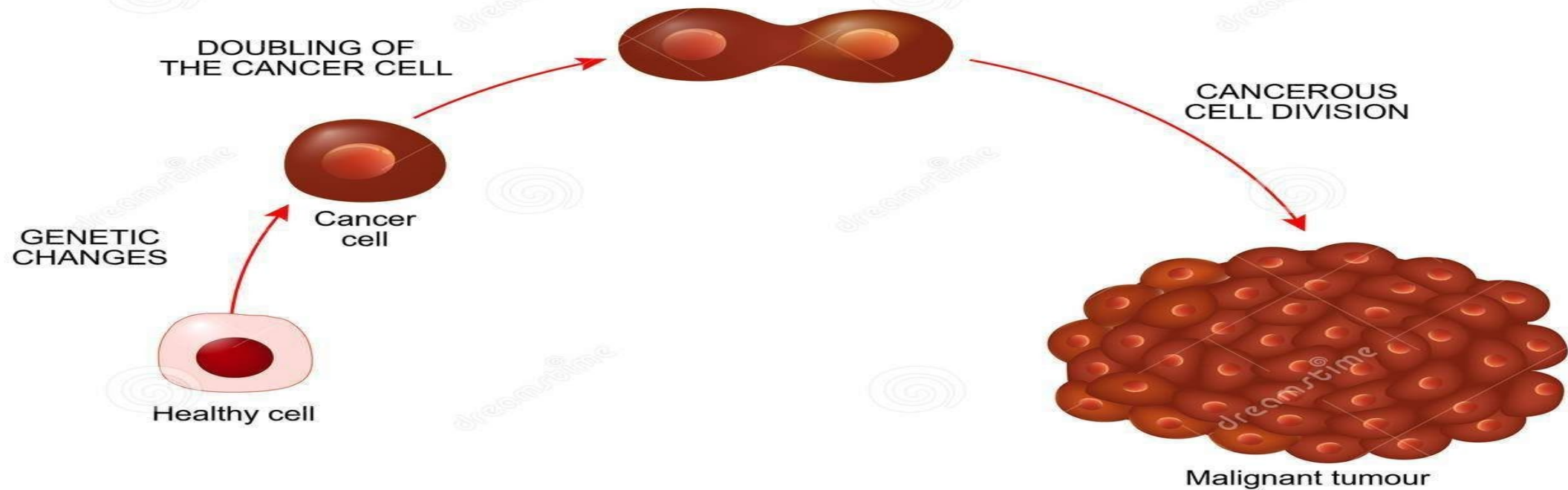
Geographic and Racial Factors

- Stomach cancer is much more prevalent in **Japan** .
- Breast cancer is much more prevalent in the United States than in Japan.
- Prostate cancer is more prevalent in African Americans than in Caucasians.

Definitions

- Neoplasia is new tissue growth that is unregulated, irreversible and monoclonal.
- Distinguishes it from hyperplasia and repair

CANCER development



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Hamartomas

- masses composed of disorganized benign cells and tissues that are indigenous to the organ/site where they arise
- Usually present from birth
- Line of demarcation between a hamartoma & a benign neoplasm is variously interpreted

E.g., pulmonary chondroid hamartoma

Choristoma (heterotopia)

- Normal tissues in abnormal locations
 - e.g. pancreatic tissue in the submucosa of the GIT
- Adrenal cells in kidney, lungs, ovaries, etc.
- Confused clinically with neoplasms, rarely sites of origin of true neoplasms

- Neoplasm may be **benign** or **malignant**.
 - Benign- Localized and don't metastasize
 - Malignant- invade locally and have the potential to metastasize
- Some tumors fall in between the spectrum of benign and malignant.

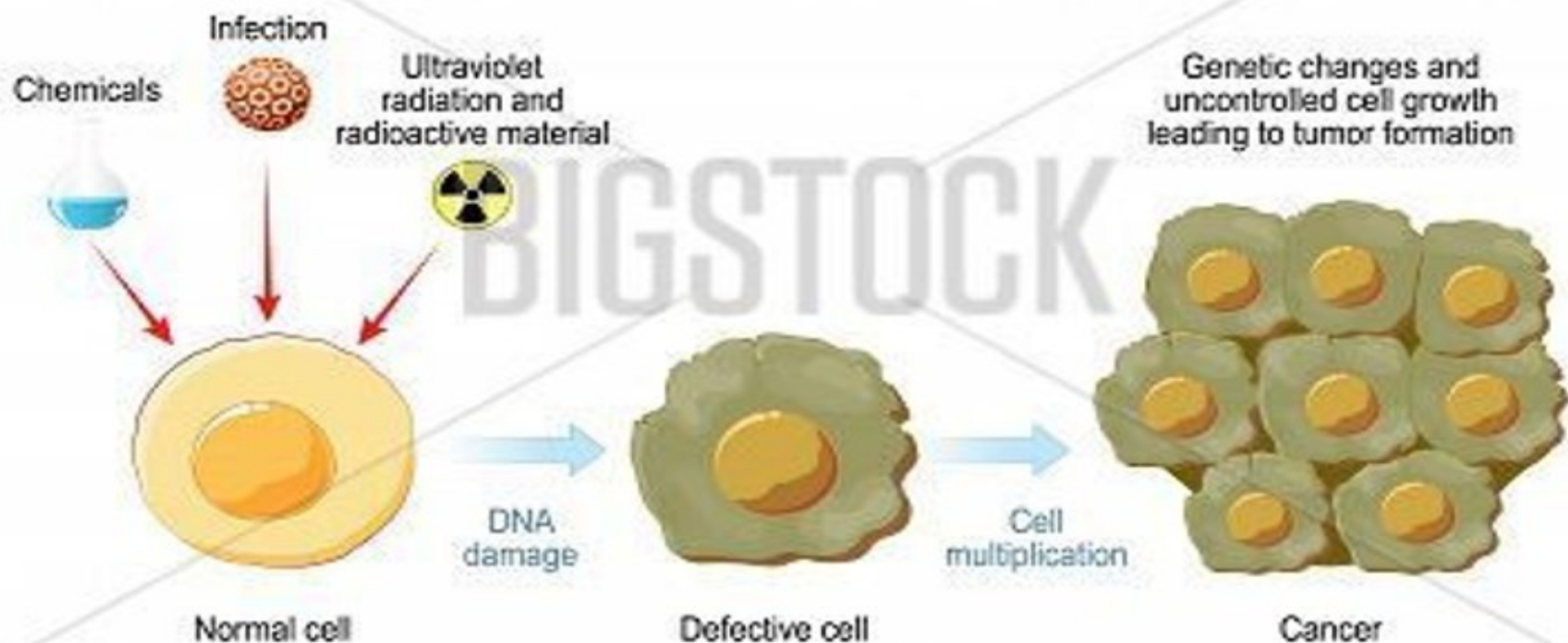
Examples of tumor nomenclature

LINEAGE OF DIFFERENTIATION	BENIGN	MALIGNANT (CANCER)
Epithelium	Adenoma	Adenocarcinoma
	Papilloma	Papillary carcinoma
Mesenchyme	Lipoma	Liposarcoma
Lymphocyte	{Does not exist}	Lymphoma / Leukemia
Melanocyte	Nevus (mole)	Melanoma

Carcinogenesis

- Cancer formation is initiated by damage to DNA of stem cells.
- The damage overcomes DNA repair mechanism
- **Carcinogen**- agents that damage DNA which increases the risk for cancer
 - Chemical carcinogens
 - Oncogenic viruses
 - Radiation

Process of cancer development



CARCINOGENIC AGENT	ASSOCIATED CANCER	COMMENTS
CHEMICALS		
Aflatoxins	Hepatocellular carcinoma	Derived from <i>Aspergillus</i> , which can contaminate stored grains
Alkylating agents	Leukemia/lymphoma	Side effect of chemotherapy
Alcohol	Squamous cell carcinoma of oropharynx and upper esophagus, pancreatic carcinoma, and hepatocellular carcinoma	
Arsenic	Squamous cell carcinoma of skin, lung cancer, and angiosarcoma of liver	Arsenic is present in cigarette smoke.
Asbestos	Lung carcinoma and mesothelioma	Exposure to asbestos is more likely to lead to lung cancer than mesothelioma.
Cigarette smoke	Carcinoma of oropharynx, esophagus, lung, kidney, and bladder	Most common carcinogen worldwide; polycyclic hydrocarbons are particularly carcinogenic.
Nitrosamines	Stomach carcinoma	Found in smoked foods; responsible for high rate of stomach carcinoma in Japan
Naphthylamine	Urothelial carcinoma of bladder	Derived from cigarette smoke
Vinyl chloride	Angiosarcoma of liver	Occupational exposure; used to make polyvinyl chloride (PVC) for use in pipes
Nickel, chromium, beryllium, or silica	Lung carcinoma	Occupational exposure

ONCOGENIC VIRUSES

EBV

Nasopharyngeal carcinoma, Burkitt lymphoma, and CNS lymphoma in AIDS

HHV-8

Kaposi sarcoma

HBV and HCV

Hepatocellular carcinoma

HTLV-1

Adult T-cell leukemia/lymphoma

High-risk HPV (e.g.,
subtypes 16, 18, 31, 33)

Squamous cell carcinoma of vulva, vagina, anus, and cervix; adenocarcinoma of cervix

RADIATION

Ionizing (nuclear reactor accidents and radiotherapy)

AML, CML, and papillary carcinoma of the thyroid

Generates hydroxyl free radicals

Non Ionizing (UVB sunlight is most common source)

Basal cell carcinoma, squamous cell carcinoma, and melanoma of skin

Results in formation of pyrimidine dimers in DNA, which are normally excised by restriction endonuclease

Activate Windows

2. Chronic Inflammation

- Persistent chronic inflammation promotes cancer.
- The link between colitis and colorectal carcinoma is an example

Table 6.3 Chronic Inflammatory States and Cancer

Pathologic Condition	Associated Neoplasm(s)	Etiologic Agent
Asbestosis, silicosis	Mesothelioma, lung carcinoma	Asbestos fibers, silica particles
Inflammatory bowel disease	Colorectal carcinoma	
Lichen sclerosis	Vulvar squamous cell carcinoma	
Pancreatitis	Pancreatic carcinoma	Alcoholism, germ line mutations (e.g., in the trypsinogen gene)
Chronic cholecystitis	Gallbladder cancer	Bile acids, bacteria, gallbladder stones
Reflux esophagitis, Barrett esophagus	Esophageal carcinoma	Gastric acid
Sjögren syndrome, Hashimoto thyroiditis	MALT lymphoma	
Opisthorchis, cholangitis	Cholangiocarcinoma, colon carcinoma	Liver flukes (<i>Opisthorchis viverrini</i>)
Gastritis/ulcers	Gastric adenocarcinoma, MALT lymphoma	<i>Helicobacter pylori</i>
Hepatitis	Hepatocellular carcinoma	Hepatitis B and/or C virus
Osteomyelitis	Carcinoma in draining sinuses	Bacterial infection
Chronic cervicitis	Cervical carcinoma	Human papillomavirus
Chronic cystitis	Bladder carcinoma	Schistosomiasis

CANCER BIOLOGY

A. Carcinogenesis

- Development of all human cancers appears to require the accumulation of **multiple genetic changes**.
- These changes can involve **inherited germline mutations** or **acquired mutations**.
- monoclonal expansion of the cell's line can cause a tumor.

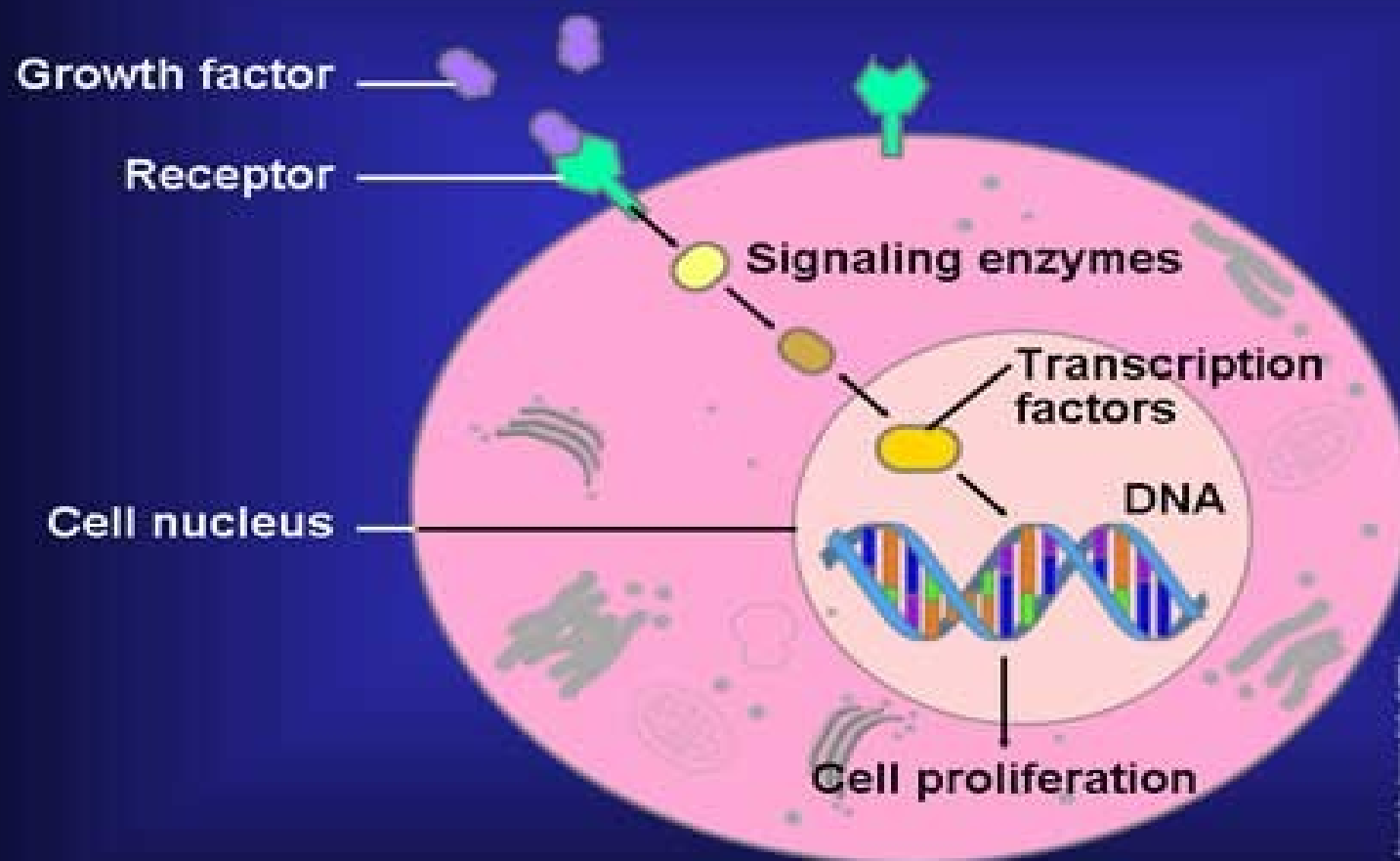
- The most important mutations in tumorigenesis involve:
 - growth promoting genes (**proto-oncogenes**),
 - growth inhibiting tumor **suppressor genes**,
 - the genes regulating **apoptosis** and **senescence**, and
 - **DNA repair** genes

Proto-oncogenes

- normal cellular genes involved with growth and cellular differentiation.
- Mutation to **oncogenes** lead to unregulated cellular growth.
- Activated oncogenes lack regulatory control and are overexpressed, resulting in **unregulated cellular proliferation**

Proto-Oncogenes and Normal Cell Growth

Normal Growth-Control Pathway

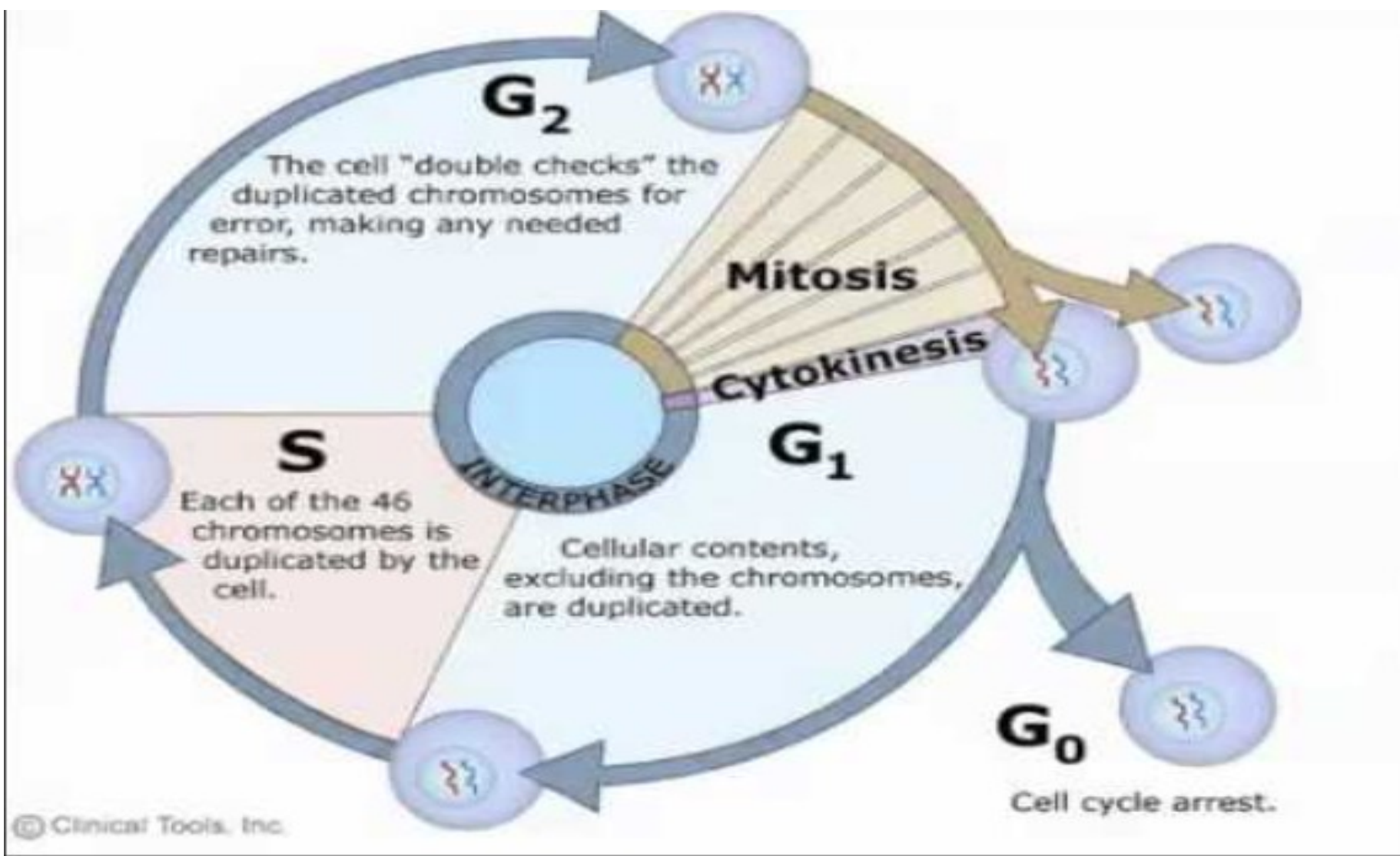


Adapted by James Folly © 2004

- Categories of oncogenes include
 - growth factors,
 - growth factor receptors,
 - Signal transducers,
 - nuclear regulators, and
 - cell cycle regulators

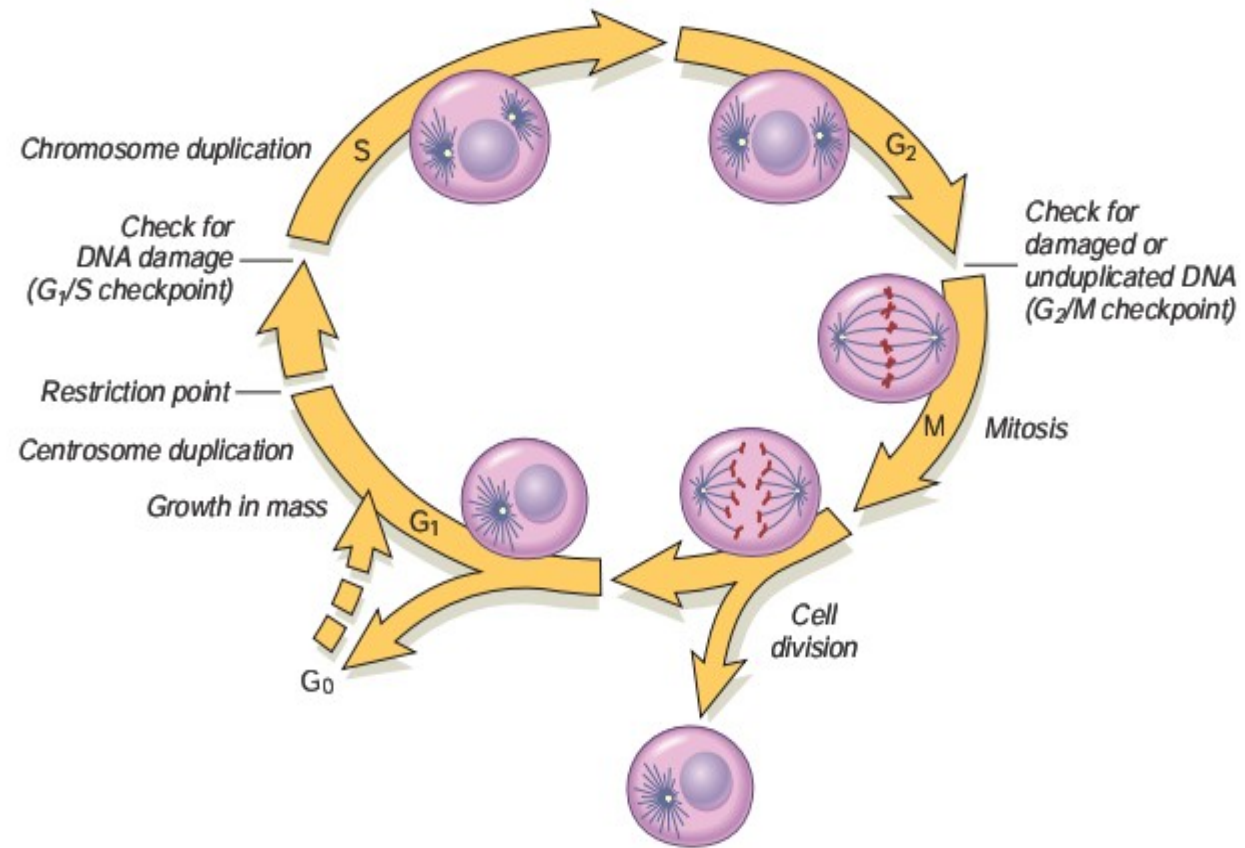
Tumor suppressor genes

- Encode proteins that regulate and suppress cell proliferation by inhibiting progression of the cell through the cell cycle.
- Oncogenes encode proteins that promote cell growth, the products of tumor suppressor genes apply brakes to cell proliferation.



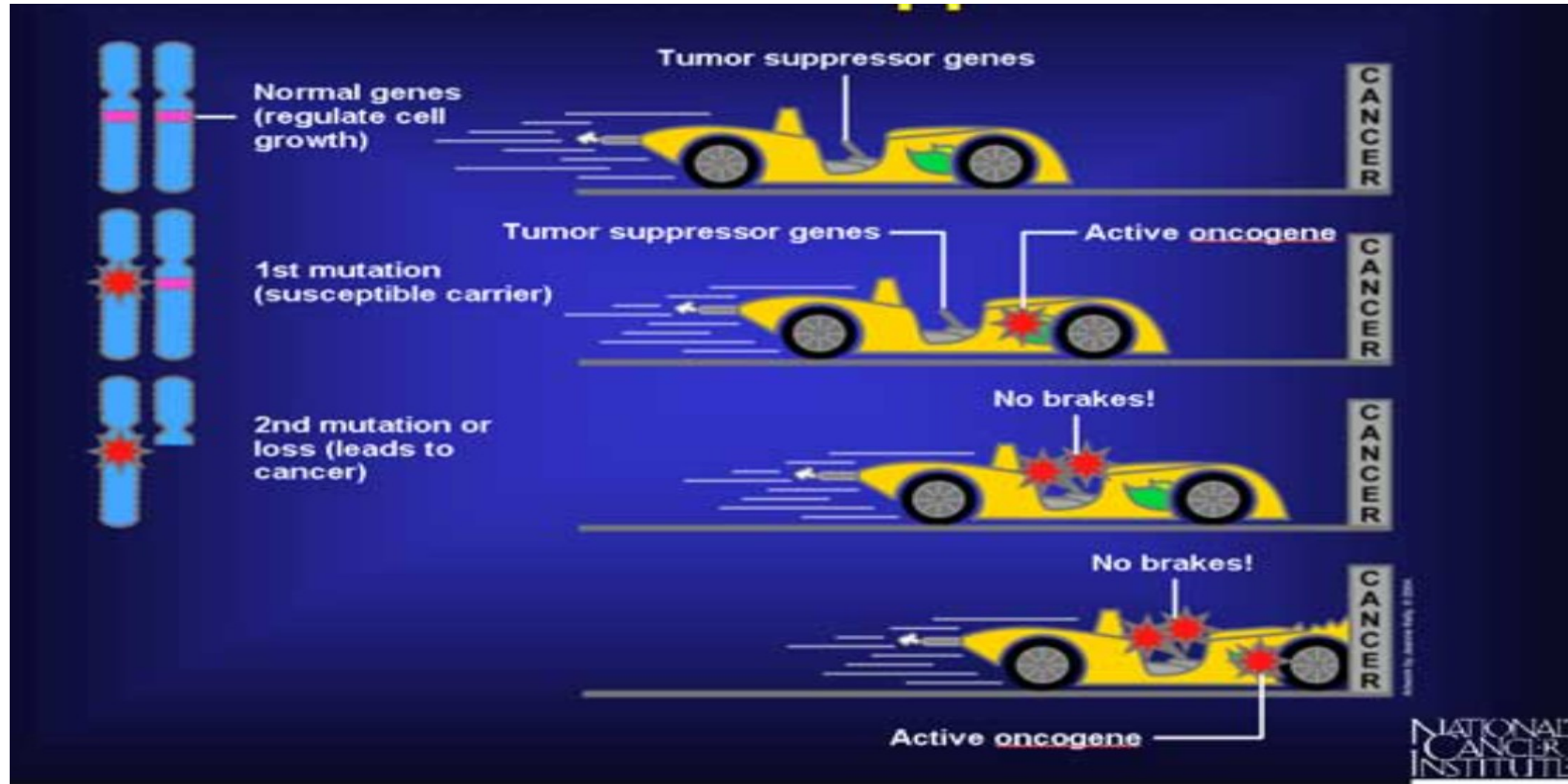
The mechanism of action of tumor suppressor genes

- **p53** prevents a cell with damaged DNA from entering **S-phase**,
- **p53** is activated by stresses such as DNA damage and assists in DNA repair by causing G1 arrest and inducing the expression of DNA repair genes
 - activation of temporary cell cycle arrest (termed *quiescence*),
 - induction of permanent cell cycle arrest (termed *senescence*),
 - triggering of programmed cell death (termed *apoptosis*).
- **Rb** prevents the cell from entering **S-phase** until the appropriate growth signals are present



KNUDSON TWO HIT HYPOTHESIS

Both copies of tumor suppressors must be knocked out for tumor formation



Regulation of apoptosis

- In addition to the action of oncogenes and inactivation of tumor suppressor genes, accumulation of neoplastic cells may result from mutations in the genes that **regulate apoptosis**

- **BAX** and **BAK** proteins: pro-apoptotic (**A**ctivate apoptosis)
- **BCL2** and **BCL-XL**: anti-apoptotic (**C**lose apoptosis)
- **BH3-only proteins** (**B**AD, **B**ID) regulate the **B**alance between pro- and anti-apoptotic signals
- BH3-only proteins sense death-inducing stimuli and promote apoptosis by neutralizing the actions of BCL2 and BCL-XL

DNA repair

- Mutations of DNA repair genes cause genetic instability.
- For example, **BRCA1** and **BRCA2** proteins are important for **double-strand break repair**.
 - Linked to breast and ovarian cancer susceptibility.
- DNA repair mutations are usually inherited.
- Diseases associated with DNA repair include **xeroderma pigmentosum** and **hereditary nonpolyposis colorectal cancer**.

Clinical features

- **Benign tumors** tend to be slow growing, well circumscribed, distinct, and mobile.
- **Malignant tumors** are usually rapid growing, poorly circumscribed, infiltrative, and fixed to surrounding tissues and local structures.
- **Biopsy** or **excision** is generally required before a tumor can be classified as benign or malignant with certainty.
- Some benign tumors can grow in a malignant-like fashion, and some malignant tumors can grow in a benign-like fashion.

Microscopic features

➤ **Benign tumors** are usually **well differentiated**, Characteristics include

1. Organized growth
2. Uniform nuclei
3. Low nuclear to cytoplasmic ratio
4. Minimal mitotic activity
5. Lack of invasion (of basement membrane or local tissue)
6. No metastatic potential

Benign neoplasm



Lipoma



Pleomorphic adenoma

➤ **Malignant tumors** are classically **poorly differentiated**,
Characteristics include

1. Disorganized growth (loss of polarity)
2. Nuclear pleomorphism and hyperchromatic
3. High nuclear to cytoplasmic ratio
4. High mitotic activity with atypical mitosis
5. Invasion (through basement membrane or into local tissue)

Malignant neoplasm or Cancer

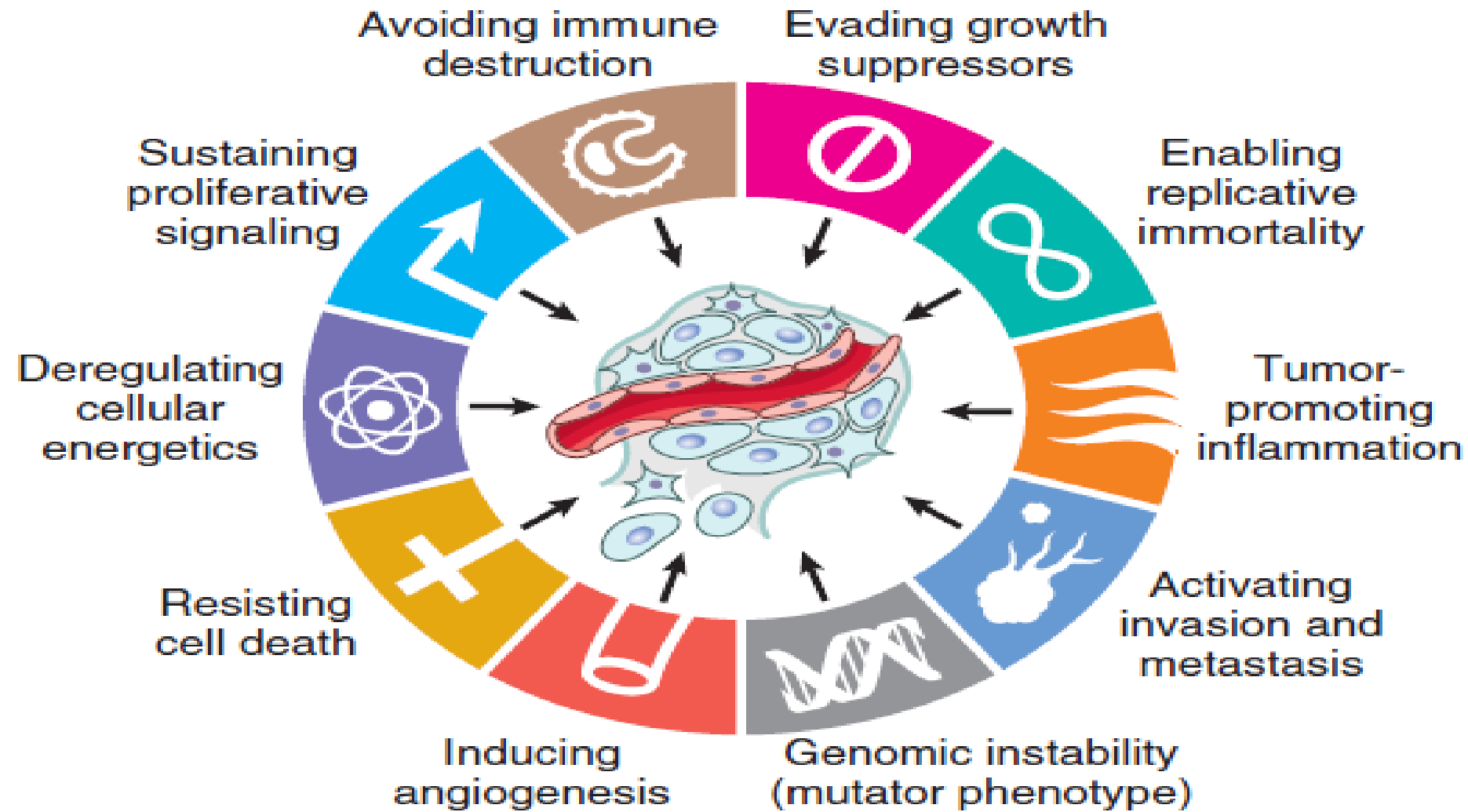


- **Metastatic potential** is the hallmark of **malignancy**—benign tumors never metastasize.
- **Immunohistochemistry** is used to characterize tumors that are difficult to classify on histology

- three fundamental features by which most benign and malignant tumors can be distinguished:
 - **Differentiation and anaplasia**, -
 - **local invasion**, and
 - **Metastasis**- Is the spread of a malignant tumor beyond its tissue of origin.
 - Lymphatic spread is the most common initial route of spread for **carcinomas**.
 - Early hematogenous spread is typically seen with most sarcomas
 - **BENIGN TUMORS NEVER METASTASIZE.**

Hallmarks of Cancer

- *Self-sufficiency in growth signals*
- *Insensitivity to growth-inhibitory signals*
- *Altered cellular metabolism*
- *Evasion of apoptosis*
- *Limitless replicative potential (immortality)*
- *Sustained angiogenesis*
- *Invasion and metastasis*
- *Evasion of immune surveillance*



Microscopic Diagnosis of Cancer

- Microscopic examination of tissue (histology) or cells (cytology) is required to make the diagnosis of cancer.

- Several sampling approaches are available:

1. Excisional or incisional biopsy

- **Excisional**: when the **entire** tumor is removed
- **Incisional**: If **only a portion** of the tumor is removed

2. Cytologic smears

Fine needle aspiration (FNAC)

PAP (**P**apanicolaou) smear

Fluid cytology

Fine needle aspiration

- The procedure involves aspirating cell and attendant fluid with a small needle followed by cytologic examination of the stained smear
- This method is used most commonly for the assessment of **readily palpable lesions** such as breasts, thyroid and lymph nodes etc.

PAP smear

- Used for the discovery of carcinoma of the **cervix**, it also detect cervical cancers at an **in situ stage**.

Fluid cytology

- Identification of tumor cell in abdominal, pleural, joint and cerebrospinal fluids

3. Advanced techniques

Immunocytochemistry (IHC)

Flow cytometry

Tumor markers

Tumor markers

- Tumor markers are biochemical indicators of the presence of a tumor.
- They include cell surface antigens, cytoplasmic proteins, enzymes and hormones.
- Tumor markers can not be construed as primary modalities for the diagnosis of cancer and thus, act as **supportive** laboratory tests.

Flow cytometry

- Identification of cell surface antigens by flow cytometry is widely used in the classification of **leukemia's** and **lymphomas**

Serum tumor markers

- Proteins released by tumor into serum (e.g., **PSA**)
- Useful for screening, monitoring response to treatment, and monitoring recurrence
- Elevated levels require tissue biopsy for diagnosis of carcinoma (e.g., biopsy of **prostate** with elevated **PSA**).

Grading and Staging of Cancer

- **Tumor grade** is a histologic estimate of the malignancy of a tumor, and
 - Differentiation and mitosis
- **Tumor stage** is a clinical estimate of the extent of tumor spread.
 - TNM staging system criteria are used for most tumor types:

- T indicates the size/involvement of the primary tumor.
- N indicates extent of regional lymph node spread.
- M indicates the presence or absence of metastatic disease.

N.B -In general, staging is a better predictor of prognosis than tumor grade.

Simplified staging

- Stage 1:- confined to site of origin
- Stage 2:- cancer is locally invasive
- Stage 3:- cancer has spread to regional structures
- Stage 4:- cancer has spread to distant sites

Cancer Prevention

➤ lifestyle Modification

- Cessation of smoking and alcohol
- Increased intake of fiber
- Weight loss
- Avoidance of environmental carcinogens

➤ **Vaccination**

- Hepatitis B virus (HBV) vaccine decreases the risk of hepatocellular carcinoma.
- Human Papillomavirus (HPV) vaccine decreases the risk of cervical cancer.

➤ Screening

- Pap smear
- Colonoscopy
- Mammography
- Prostate-specific antigen

➤ **Preemptive Treatment for Precursor Conditions**

- Treatment for regimen for **Helicobacter pylori** may prevent malignancies such as **mucosa-associated lymphoid tissue cancers**.
- Treatment for **gastroesophageal reflux disease (GERD)** may prevent the development of **adenocarcinoma of the distal esophagus**.

References

- Robbins and Cotran Pathologic Basics 10th edition

Thank you!!!